DECO3500 SOCIAL & MOBILE COMPUTING · SEMESTER 2, 2026 · DESIGN PROJECT

Space-Cap

A group time-capsule ritual, supporting relationships through nostalgia and reminiscence

Team: Team 12, NO_PARKING(!)

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GitHub repository: https://github.com/batinion/DECO_3500
Project wiki: https://github.com/batinion/DECO_3500/wiki

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2. Ethical Considerations (wiki page export)

Design Process Overview

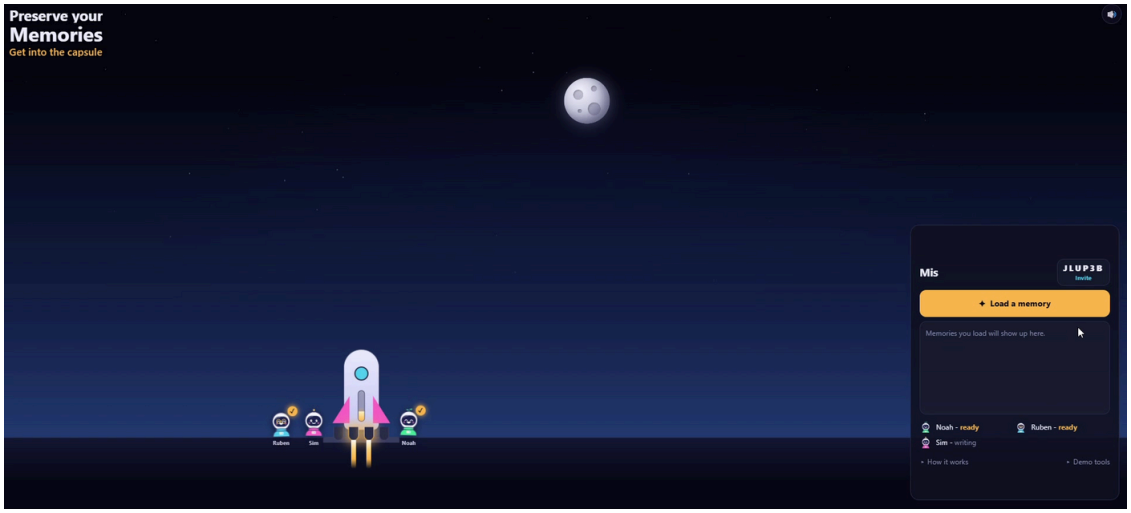
Poster: Team 12 poster (PDF) · **Promotional material:** PDF

Prototype: Space-Cap, final build. No login: one person starts a crew and shares the code. Setup instructions: prototype README.

Supporting material: Ethical Considerations · Prototypes and Evaluation · User testing · Design artifacts · Miro board

Summary

People keep memories, but almost always alone, and every similar product seals a capsule that one person writes and one person opens. *Space-Cap* makes remembering something a group does. A crew of up to six writes prompted memories about each other across a semester and seals them in a rocket. Years later, in pretend, they summon it home, fly one shared ship to rescue the scattered memories, and open the capsule together. Nobody can do either alone.

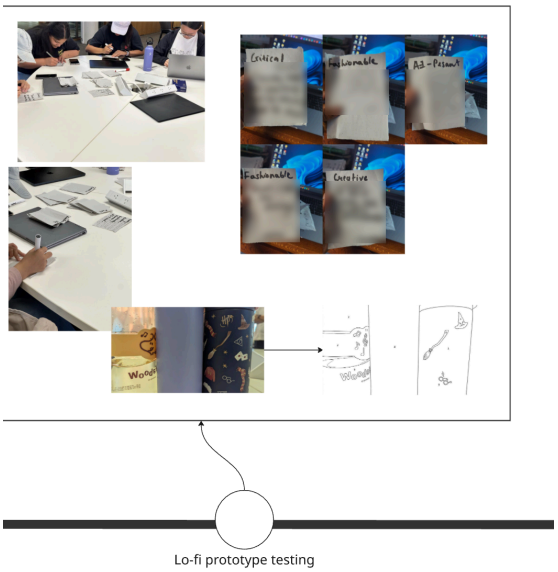


No host: everyone joins one crew code.

Process

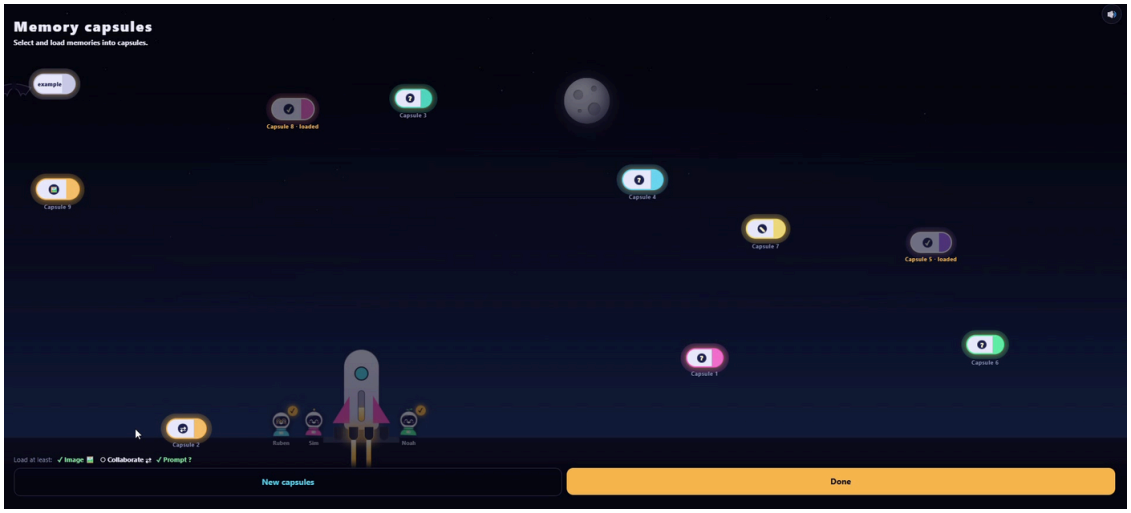
Research. The idea began with the time capsule in *MacGyver*. Literature showed that remembering serves social needs and identity (Bluck et al., 2005; Conway, 2005) and that digital mementos feel less significant than physical ones (Petrelli & Whittaker, 2010). Ten interviews by three of us gave six themes; two shaped the design. People curate only when forced, and keeping a memory is not revisiting it. So memories come from prompts, not an open feed, and the capsule stays sealed on purpose.

The lo-fi chit activity. Our first test was on paper. Classmates wrote notes about each other on trait chits, then coloured a drawing together before the notes were handed over. They wrote freely and wanted to know who wrote what, which gave us memories about each other and guessing the author. The colouring confused them, and it failed again in our first build.



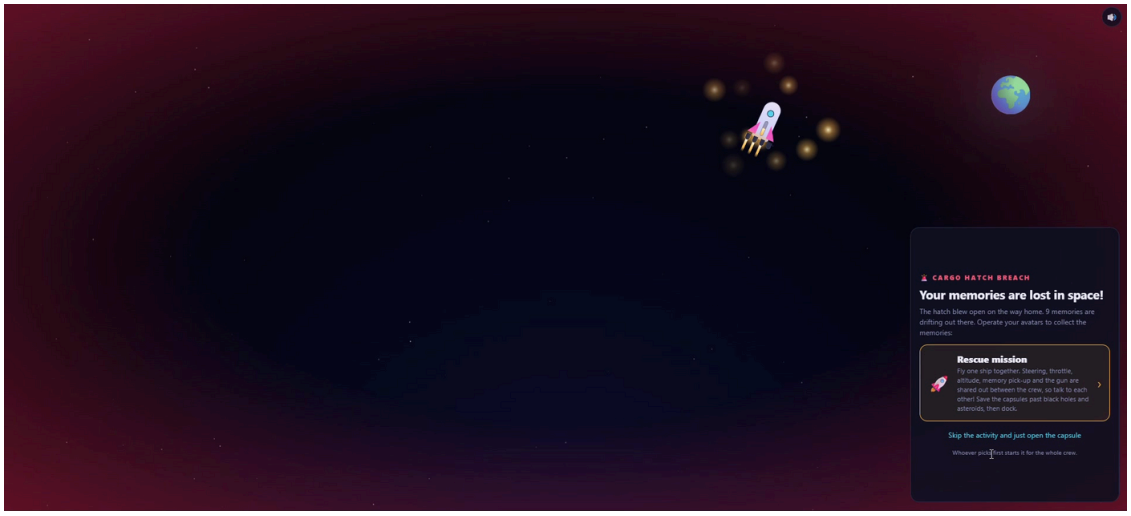
Trait chits and the colour-in drawing.

First build. Prototype v1 ran on a shared Mission Control screen. Testers missed that the prompts were about each other, felt the host screen made them followers (*"group experiences are rarely leader-follower"*), and found one-sitting collection felt like a classroom exercise. We removed the host, so everyone uses their own device, and spread collection across a semester with 33 course-week prompts. A full run then showed that years passing in space made no sense, so time now passes on Earth.



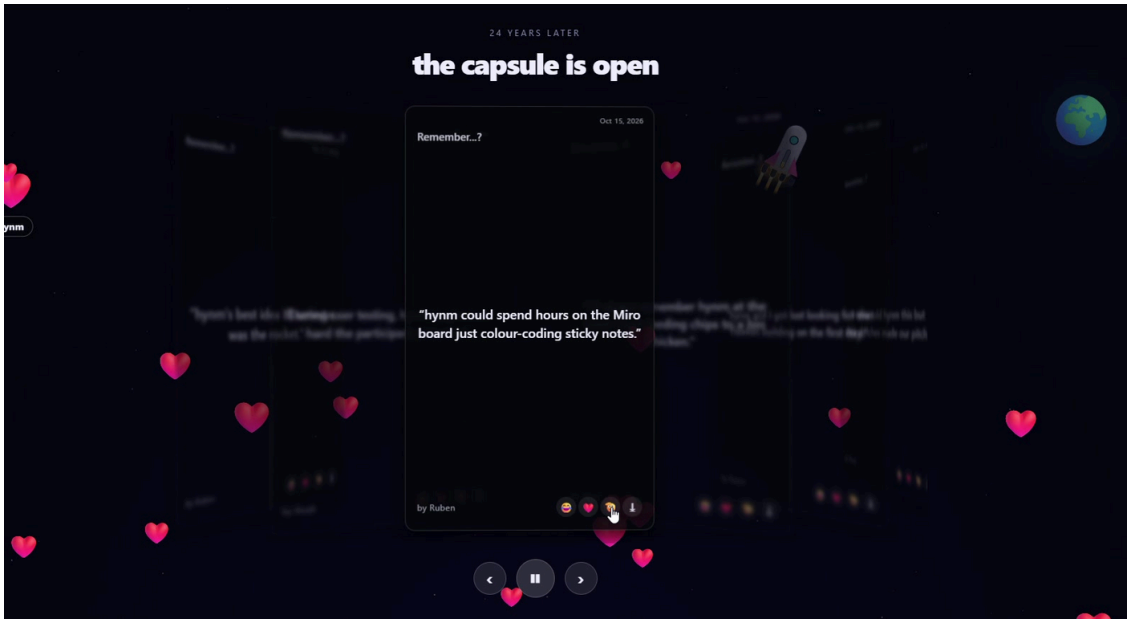
Capsules fill across the semester without showing what is inside.

Finding the unlock. Rather than argue, nine participants in three crews compared a colour game, one ship each, and one shared ship. The colour game was clear (mean 4.5 of 5) but least collaborative (3.0), so we dropped it. The shared ship matched it on clarity and led on working together (4.5), though three of its four ratings came from one crew. Players kept talking (*"who has the gun?"*), but one-word answers split between *"Cooperation"* and *"Fun but confusing"*: nobody realised the five controls were split.



The brief names the controls but not who holds each.

The full run. The reveal held up: *"I learnt something about how my crewmates see me"* averaged 4.3. Our weakest result sits under our main claim: *"I would add something next week without a reminder"* had a median of 3. We kept the bounded design and accept that cost.



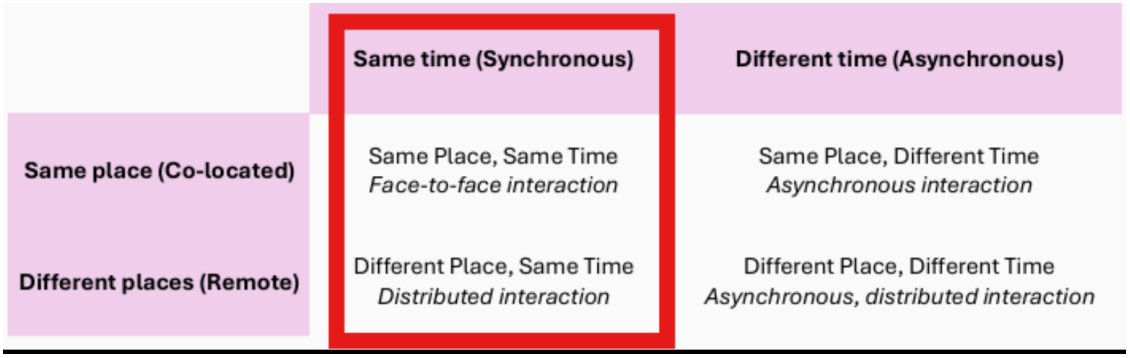
Memories open one at a time; reactions appear on every screen.

The tradeshow. The shared ship worked because we said aloud that the controls were split, so the briefing did what the interface cannot. Visitors also wrote beside the people they wrote about, so the reveal held no surprise. The design's value is in collecting apart and opening together.

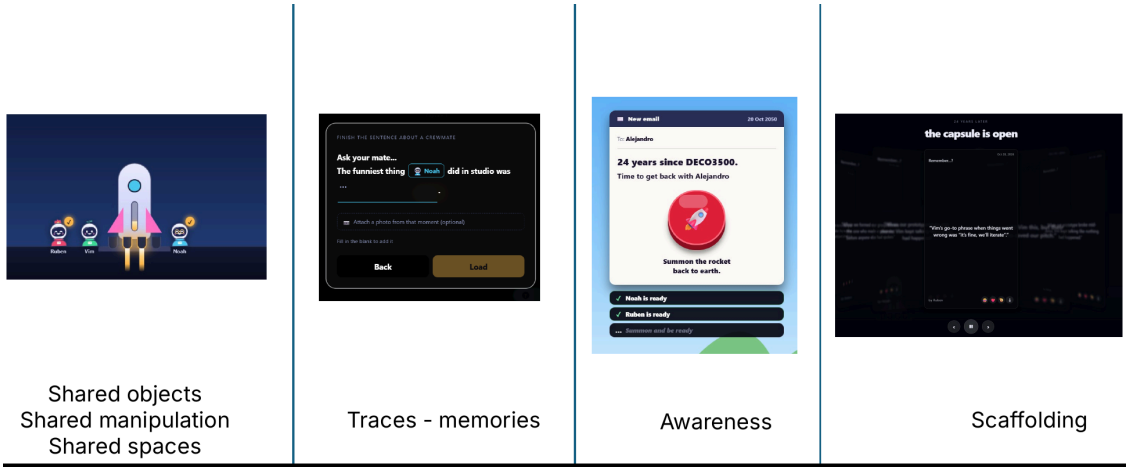
Stage	How	What we learnt
Lo-fi chit activity	Paper chits, then group colouring	People write openly and want to know the author; colouring confuses
Prototype test 1	Two groups, Mission Control build	Prompts and theme worked; host screen, one sitting and colouring did not
Narrative run	Full rebuilt flow	Time should pass on Earth, not in space
Prompt activity	Prompts in text, voice and images	Prompts aid recall; let one memory name several people

Stage	How	What we learnt
Test 2: unlocks	Nine participants, three crews	Shared ship most collaborative; controls not self-explaining
Test 2: full flow	Six participants, start to finish	Reveal strong; unprompted contribution weak; glitches with five players
Tradeshow	About fifteen visitors	Works when the split is explained aloud

Theory. On the time-space matrix, collecting is different-time, different-place, which suits digital media (Hollan & Stornetta, 1992); opening is same-time. Shared activity does more for people together than awareness alone (Olsson et al., 2020), so the unlock is a task, not a countdown. Groupware fails when one person works and others benefit (Grudin, 1994), so there is no host and every player holds a control.

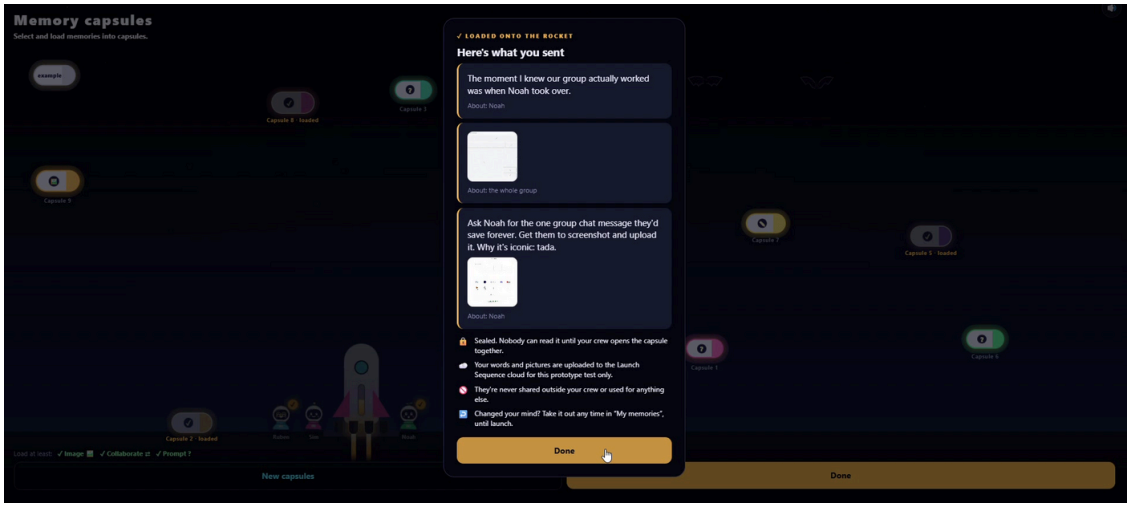


Opening sits in the same-time column.



Groupware elements in the final build.

Ethics. The capsule holds memories about people who did not choose what was written. Our Ethical Considerations page maps fourteen risks; the privacy card is our main consent surface.



The privacy card after each memory.

Limits. The years pass only in pretend. Reopening across time zones is designed but not built, and a crewmate who has died raises a question we have not answered. The largest limit is the future itself: a capsule sealed today must open in 2050 on technology the solution cannot anticipate. We tested only with students.

Who did what

Member	Contribution
Dev	Poster and promotional material; time-jump design and opening sequence
Vim	Built the prototype, first version to final build; documentation merge; three interviews; meeting records
Shirish	Research lead: five interviews; the no-host, semester-long rebuild
Dattatray	Repository and wiki; documentation; testing plan and evaluation
Dong	Two interviews; Guess Who reveal; final build contributions

References

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Ethical Considerations

We used the **Ethical Disclaimer** canvas from ethicsfordesigners.com. It is a living document, so we revisit this page after each round of testing and before the tradeshow. This version covers the final prototype.

Our tutor told us that protection, ethics, privacy and data handling must be properly justified, not just ticked off. So for every risk we say what we have **done**, what we **plan** to do before the tradeshow, and what is **outside our control**.

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1. Current situation

Classmates and friends scatter at the end of something shared. Their memories end up spread across camera rolls and group chats. Existing time-capsule products are made for one person writing to their future self.

Space-Cap, called *Launch Sequence* in our code, is a group ritual. A crew of up to six people adds short memories about each other **over a whole semester**: written prompts about the course, photos and sketches, selfies and voice notes. The memories stay sealed. When everyone is ready, the rocket launches and the crew is carried forward 24 years to 2050 (this part is pretend). The crew then flies one shared spaceship to rescue the scattered memories and opens the capsule together.

2. Stakeholders (who is affected)

Who	How they are affected
Crew members who write	Want a good way to mark the end of something shared. They also give us their face (selfies) and their voice (voice notes)
Crew members who are written about	Every memory is about someone else. They didn't choose what is said about them, and the whole group sees it with the author's name on it
People in photos and screenshots	May not be in the crew and haven't agreed to appear. A group-chat screenshot also shows other people's names and messages
Classmates named in memories who aren't there	At the tradeshow, visitors write about classmates who are not at the stall
Whoever runs the server	Their laptop holds every crew's memories, photos, selfies and voice recordings
Test participants and tradeshow visitors	Give us real memories about real people. Test sessions were also recorded on audio
Players with different abilities and devices	The shared game needs keyboard (or touch) control, quick reactions, and attention to sound and visuals
Creators of material we borrow	The time-lapse GIFs and any other outside media
The team as researchers	We are responsible for agreement, privacy and safe storage of recordings, transcripts and uploads

3. Design intentions

To bring a small group of classmates or friends closer by remembering the semester together, just as they are about to go separate ways, and to give them a reason to meet again.

It is not meant for strangers, for groups bigger than six, for groups in an unresolved conflict, or for children (we assume adult university students). It should never be used against someone.

4. What could go wrong

#	What could go wrong	Who is affected	Where it comes from
U1	**A private secret or hurtful comment is shown to the group.** Someone writes a confession or a criticism about a friend, and it appears in the reveal with both names on it	The person written about	Prompts like *"I never told ___ this, but ___"* and *"The tutor feedback that hit ___ hardest was ___"*. The author is always shown in the reveal
U2	**A joke goes wrong.** A teasing prompt is filled in unkindly, and the person sees it for the first time in front of everyone	The person written about	Prompts like *"never let ___ near ___"*, *"Nobody could get ___ to stop ___"* and the "worst photo" prompt
U3	**Someone is hardly written about,** and the reveal makes that obvious	The person written about	Writers choose who each memory is about. Nothing makes sure everyone is covered
U4	**The reveal shows how people really saw someone,** which they didn't ask for. One tester liked this idea for a new sports team, but it can also hurt	The person written about	The reveal shows every memory and every author
U5	**Photos and screenshots of people who never agreed.** For example, a group photo with someone outside the crew, or a group-chat screenshot showing other people's names and messages	People in photos and chats	Photo attachments, and prompts that ask for a photo together or a chat screenshot
U6	**Memories stay on someone's laptop for ever, and the app describes this misleadingly.** Crews, memories and uploaded files stay on disk. Removing a memory deletes its record but not its file. The privacy card says "cloud" and "never used for anything else", but everything is on the laptop running the server	Everyone	`server/data/crews.json`, the submissions folder, and the privacy card wording
U7	**A promise we can't keep.** The experience suggests years of safekeeping, but the wait is pretend and the prototype only keeps data on one laptop for a semester	Crew members	The time-jump story
U8	**People lose access to their own memories.** A tester said: *"Three years is a long time. People get hacked. How do you protect access, and how do I keep access? I don't want to lose the memories or miss out."*	Crew members	No accounts and no way to recover access

#	What could go wrong	Who is affected	Where it comes from
U9	**Research and the tradeshow without clear agreement.** Testers and visitors write real memories about real people, and some of those people aren't there	Testers, visitors and the classmates named	Our testing sessions and the stall
U10	**Faces and voices identify people more than text does.** Selfies and voice notes (up to 60 seconds) are stored as files. A recording can pick up other people's voices in the background	Crew members and people nearby	Selfie and voice-note capsules
U11	**One person holds the crew back, or feels pressured.** The crew can't launch until everyone has added three memories and tapped Ready. It can't come home until everyone presses "Summon the rocket". There is no way to leave a crew	Slow or unwilling members	The ready and summon steps
U12	**Some players can't join in fully in the shared game.** There is one ship and five controls, shared out at random. A player may get a control that is hard for them, such as one that needs fast timing	Players with different abilities	The shared spaceship game
U13	**Borrowed media with unclear rights.** The time-lapse GIFs came from outside sources, and some files in the repository are named after TV shows	Original creators	`server/public/future/`
U14	**Weak security habits.** The camera and microphone need a secure (HTTPS) page with a home-made certificate. This shows a "not private" warning that people get used to clicking through. The certificate's private key is also stored in the repository	Everyone	`server/certs/`

5. What we will do about it

Risk	What we have done and will do	Outside our control
U1, U2: hurtful or private content	**Done:** we rewrote the prompts around the shared course journey, so most ask for funny or good course moments. At the tradeshow, visitors are asked for joyful memories. **Planned:** remove or rewrite the four prompts that still invite confessions, teasing or criticism (*"I never told ___ this, but ___"*, *"never let ___ near ___"*, *"Nobody could get ___ to stop ___"*, *"The tutor feedback that hit ___ hardest was ___"*). Rethink the "worst photo" prompt. Add a clear "keep it kind" line before writing. **Considering:** letting the person written about preview and skip their own card before the group reveal	What friends say to each other outside the app

Risk	What we have done and will do	Outside our control
U3: someone left out	**Done:** the reveal doesn't rank people (awards are no longer shown) *(confirm)*. **Considering:** asking each writer to cover every crewmate at least once	Existing group dynamics
U4: unwanted exposure	**Done:** we present it as remembering together among people who already know each other, not as a judgement of anyone	How someone feels about honest feedback
U5: photos and screenshots of others	**Planned:** a clear line at upload: only add photos or screenshots of people who are here and happy to appear, and hide or crop other people's names	Photos taken outside the app
U6: stored data and misleading wording	**Done:** our README tells whoever runs the server to delete the data and submissions folders after every session. **Planned:** change the privacy card to say plainly that data is stored on the team's laptop, and to list photos, sketches, selfies and voice recordings. Delete uploaded files when a memory is removed. Delete all crew data and files after the tradeshow	Copies people keep elsewhere
U7: promise we can't keep	**Planned:** say clearly in the briefing, poster and pitch that the wait is pretend	
U8: losing access	**Considering:** treat it as a design task for the concept, with regular reminders and a way to recover access, as the tester suggested. We will document it even though we haven't built it	Whether a real service would still exist in five years
U9: clear agreement for research	**Done:** consent was collected before testing. Sessions were audio-recorded on team phones, and participants knew. Our test transcript uses first names only. At the tradeshow, visitors wrote about each other rather than about absent classmates, so the risk we had prepared for largely did not arise. **Planned:** a short, plain notice at the stall (what is stored, where, for how long, and that you can remove a memory before launch). Remove names from notes before they go in the repository. Delete the audio once it has been written up	What visitors tell friends about the stall
U10: faces and voices	**Done:** voice notes are capped at 60 seconds. **Planned:** add selfies and voice notes to the privacy card. Remind people not to record others without asking. Make sure players can skip these capsules *(confirm)*	Recordings people make elsewhere
U11: pressure and being held up	**Done:** a visible skip button lets any player bring the memories home straight away. **Considering:** a way to leave a crew, and letting the crew carry on when someone steps back	Social pressure inside the group
U12: joining in the shared game	**Done:** the crew shares control of one ship, so nobody has to do everything. Arrow keys and WASD work as well as touch. Mistakes only add seconds to the crew's time (+8 s for black holes, +3 s for asteroid crashes), and the game only ends once every memory is saved. **Considering:** letting players swap roles, and testing with a wider mix of people	Differences in ability and devices
U13: borrowed media	**Planned:** replace outside GIFs with our own or properly licensed ones, or credit the sources. Remove unused files from the public repository	

Risk	What we have done and will do	Outside our control
U14: security habits	**Planned:** remove the private key from the repository and make a new one on each machine. Explain the one-time warning at the stall	Browsers' own warning wording

6. Evidence from the semester

- **Tester feedback:** the worry about long-term access (U8) came straight from a tester.
- **UQ colour-game test:** one tester said *"No letter for me?"* while others had two or three. Another hadn't memorised part of the picture. We later scrapped the colour game. It still made us ask whether anyone can end up with nothing to do, which is why the final game gives every player a control.
- **Prototype test 2 (13 and 15 October), anonymity:** one participant expected the memories to be anonymous, and was uncomfortable that authorship was visible. They asked for anonymity as an option, not the default. Evidence for U1, U2 and U4, and it cuts against itself: anonymity removes the accountability that keeps the reveal kind.
- **Prototype test 2, the three required memories:** participants said requiring all three memory categories asked too much, because group dynamics vary and some people have only one memory worth writing. They also asked for an option not to upload at all. This is tester evidence for U11, which until now rested only on our own reasoning.
- **Prototype test 2, a crewmate who cannot be there:** one participant asked what happens when everyone must be present to unlock and a crew member is unavailable or has died. We treat this as a memorialisation question rather than a bug, and it extends U8 beyond lost passwords.
- **Prototype test 2, photographs of other people:** participants raised consent for photographs of people who had not agreed to appear, and worry about uploading something inappropriate by accident. This confirms U5 from outside the team.
- **Tradeshow (20 and 22 October), who gets written about:** six teams came through the stall and visitors wrote about each other rather than about classmates who were not present, which was our main worry under U9. Some teams wrote about the team's shared experience instead of about individuals.
- **Tradeshow, the surprise defeated:** visitors filled in prompts sitting beside the person they were writing about, and one simply asked his partner the answer. The capsule held nothing new for her. Sealing protects content from the group, not from the person sitting next to you.
- **Tutor feedback:** our tutor asked us to justify protection, ethics, privacy and data handling properly.
- **Readings:**
  - Winner (1980) says people are often left out through neglect, not bad intent, and that early design choices are hard to undo. We use this for U11 and U12.
  - Akrich (1992) shows that designers build assumptions about users into their products. Ours assumes everyone is happy to be named in a memory, enjoys group play and has a laptop that can run it.
  - Huang et al. (2026) warn that "one person's memory may be another person's trauma" and describe "a worry of being judged" on social platforms. We use this for U1 to U4.
  - Zhang et al. (2026) call for "emotional safety and user control" when people revisit memories. This is why we are considering a preview and skip for the person written about.

7. What changed because of this

Done

- The prompts are built around the journey of the course.

- Consent was collected, and sessions were recorded with agreement.
- Voice notes are capped at 60 seconds.
- The final game shares one ship across the crew and has a skip button. **Still planned**
- Remove or rewrite the four risky prompts.
- Fix the privacy card wording and add the notice at the stall.
- Delete uploaded files when a memory is removed, and delete all crew data after the tradeshow.
- Add the photo and screenshot reminder at upload.
- Replace or credit the GIFs, and remove the private key.
- Say clearly that the wait is pretend. **Not doing, and why**
- **Invented memories at the stall.** We want real memories from the course, so we reduce the risk with joyful prompts, a notice at the stall and deleting the data.
- **Real accounts and login.** The brief says not to build them. Long-term access is covered as a design task (U8).
- **A moderation system.** We reduce harm through the design instead of checking what people write.

Change log

When	Change
Earlier draft	First version using the Ethical Disclaimer canvas
Final prototype update (Oct 2026)	Updated for the final build: selfies and voice notes, the shared spaceship game, the dropped colour game, the misleading privacy wording, borrowed media and security, with a done / planned / not-doing status for every risk

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